

Creation of a coral park in the Hawaiian archipelago

The Malama Kai Center

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Coral Reefs in the Hawaiian Archipelago – Current Status



Hawaii – one of the most **isolated** archipelagos with exceptional marine biodiversity (high **endemism**)
→ **Essential role** of coral reefs: coastal protection, local economy (tourism, fisheries), and ecosystem services

Climate threats :

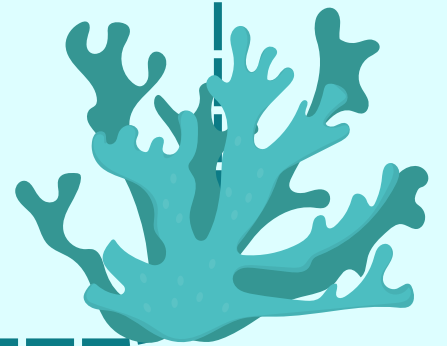
- Rising ocean temperature = coral bleaching
- Hurricanes (Iwa 1982, Iniki 1992) = structural reef damage

Recent events :

- 2014-2015 (ENSO) = mass bleaching (~68% at some sites)
- 2019 : recovery observed, but species-specific vulnerability such as *Pocillopora damicornis*, coral of the genus *Porites* + resistant

Anthropogenic factors affecting reefs:

- Coastal pollution (runoff, wastewater)
- Tourism (~7M visitors/year, 80% aquatic activities)



Coral bleaching expected to become + **frequent** and + **intense** in the coming decades



Market needs assessment

Problem statement :

→ **Thermal stress + antropogenic pressure** (tourism) = damages reefs requiring **restoration action**

Demand :

- Development of **ecotourism**
- Target audience : **tourists + local communities** dependent on marine resources
- Strength: high tourist attraction to coral reefs
- Respect for nature, a core value of Hawaiian culture



Supply & Strategy :

- Collaboration with **parteners** (associations, foundations)
- Activities (school, summer camps → awareness-raising)
- Differentiation from conventional tourist activities : snorkeling, diving (competitors) by offering a **scientific** and **educational** approach

Funding Opportunities

- *Hawai'i Conservation Program*
 - *National Fish and Wildlife Foundation (NFWF)*
 - *Hawai'i Conservation Alliance Fondation*
 - *Hawai'i Community Foundation*
- Protection, conservation & ecological restoration, climate change mitigation



Proposed product/service

Mission : Establish an innovative rehabilitation center combining **science** and **local engagement** to protect and restore coral reefs

→ **Name of the project** : **Malama Kai Center** ("*Protect the ocean*" in hawaiien)

OBJECTIVES

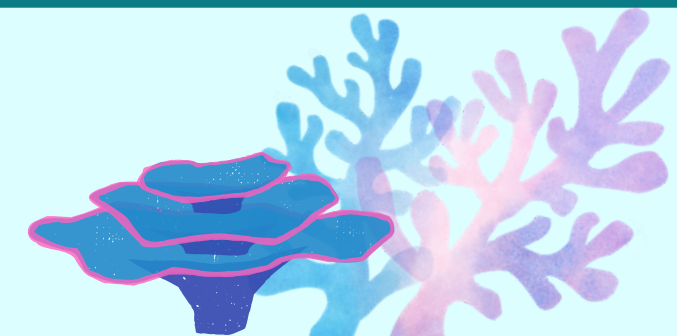
- **Raise public awareness** of the role of coral reefs for marine biodiversity and coastal communities
- **Prevent** human and climate impacts on marine ecosystems
- **Actively** rehabilitate reefs using techniques such as coral fragmenting (coral gardening)
- **Combat** climate change through sustainable solutions

Inspirational Projects:

- *Coral Gardeners* (Indian Ocean) : 100 000 corals planted
- *Blue Ventures* (Indonesia) : Fishery restauration for rural communities
- *Manta Trust* : Conservation (rays and sharks)
- *The Ocean Cleanup* : Innovative solutions to clean the oceans

Financial Support & Partnerships :

- Private donations, grants, CSR initiatives
- Ecotourism and merchandising (branded products)
- Collaborations with universities, research institutes, and academic funding



Preliminary data

Nurserie des coraux:

- Élevés durant 1 an
- Taux de survie : 92%

Replanting :

- Fixed to a damaged reef
- Survival rate : 82% à 85%

In the Hawaiian Archipelago :

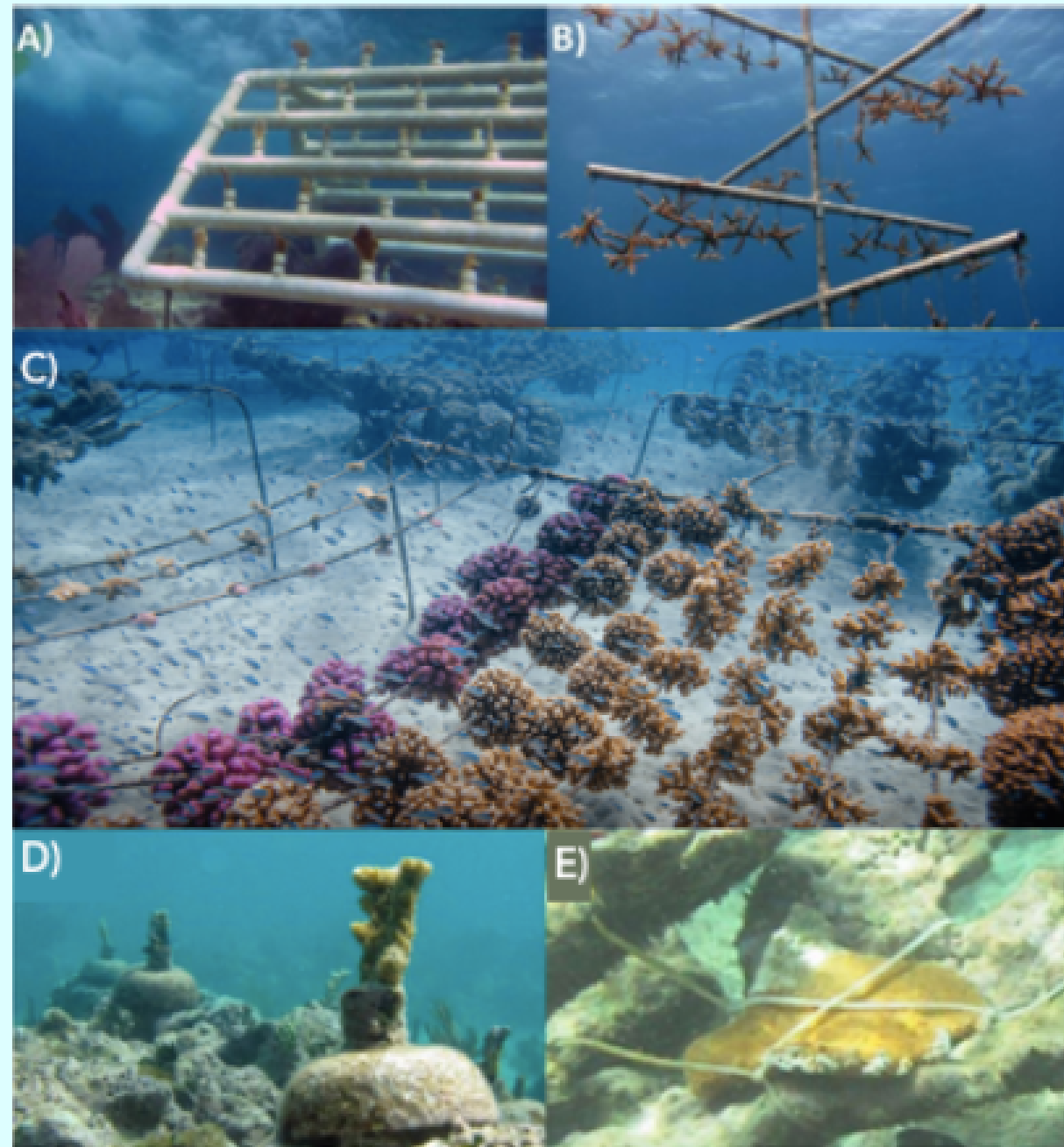
- Study conducted on Oahu
 - Multi-directional PVC system
- *M. capitata* : better survival on the “tree” (61%)
 - Higher calcification rate
 - Common species
 - Wide distribution
- *M. flabellata* : Low survival (14%)
 - Better adapted to oceanic zones

Structures :

- Systèmes en PVC
 - uni-directionnel (A)
 - multi-directionnel (B)
- Structure métallique (C)
- Support en béton (D)
- Attache en nylon (E)

- 1 fragment = 1 colonie
- Restauration ciblée

Fig 1. Structures utilisées pour le replantage de coraux



In our park :

- Test two techniques :
 - Multi-directional PVC (B)
 - Metal structures (C)

Methods are low-cost and show high survival rates

Development Plan

Project Goal : **protect** and **restore** local reefs and raise public **awareness**

STEP 1 : first 6 premiers months

- Local **collaboration** and **partnerships**
- Field studies :
 - **Inventory** of fauna and flora
 - **Assessment** of degradation
 - Ecosystem **modeling**
- **Multidisciplinary** team
 - Biologists, chimists, divers...

STEP 2 : 4 to 10 months

- **Development** of facilities and transplantation areas
 - *Hawaiian Department of Land and Natural Resources* (DLNR)
 - **Protected site** : Molokini Crater
 - Biodiversity conservation program
- Setup of necessary equipment for future activities
 - Information panels, welcome area ...
- Completion of **staff** selection

STEP 3: 9 months and more

- Welcoming the first **visitors** & launching **activities**
- **Interactive** trails, **informational** panels, awareness-raising
- **Revenue** generated will fund the coral park
- **Communication** strategy via social media:
 - Tiktok, Instagram et Youtube + online events



Analyse SWOT

STRENGTHS

- Environmentally beneficial
- Engages the public in a fun and educational way
- Project credibility reinforced by scientific involvement
- Contributes to coral protection
- Collaboration with research centers and universities
- High survival rate of replanted corals

WEAKNESSES

- Geographical location
- Dependent on tourism
- Financial challenges, not highly profitable
- Open natural environment
 - uncontrollable parameters
- Reliant on grants / donations

OPPORTUNITIES

- International partnerships
- Grants and scholarship
- ↗ demand for ecotourism
- Outreach in schools, companies, conferences...
- Merchandising development
- Promote research and knowledge

THREATS

- Climate change and natural hazards
- Local competition (dive centers, snorkeling activities)
- Unstable economic sector
- Legal and administrative complexity
 - related to protected areas
- Vulnerability to environmental disturbances
- Political situation affecting government subsidies

Budget Summary

Initial Investment

- Diving / snorkeling equipment
- Research laboratory
- Center development: educational center, landscaped outdoor area
- Marketing
 - Website, brochures, social media
 - Launch events

→ 132 000 €

Annual costs :

- Salaries : scientists, facilitators, maintenance staff
- Insurance
- Maintenances : diving equipment, ...

→ 242 000 €

Estimated Annual Revenue (Self-financing)

- Educational and interactive trails: different pricing for children, students, adults, locals
- Educational activity days
- Occasional workshops
- School outreach programs
- Research grants
- Gift shop (merchandise)

→ 237 800 €

Sales Plan

Novelty on this island → Market exclusivity + growth of ecotourism
Financial objectives: generate €250,000 in revenue during the second year

Target audience:

- General public
 - Adults, children, locals, tourists
- Specific audience
 - Scientists, schools et educational institutions

ACTIVITIES :

- Themed days (Ocean Day)
- Conferences
- Educational workshops
 - Conservation, restauration, pollution,...
- Snorkeling, diving

PRICING STRATEGY :

- Differential pricing (age, students, etc.)
- Annual subscriptions
- Special packages (school groups, families, etc...)
- On-site gift shop
 - Eco-friendly merchandise
 - bags, notebooks, pens, magnets,...

ADVERTISING :

- Social media
 - Tiktok, Instagram, Youtube,...
 - International reach
 - Real-time sharing
- Partnerships
 - Travel agencies and hotels



Thank you for your attention :)

